

Finding Appropriate Hypothesis

$f(x)$
 $\rightarrow \frac{df(x)}{dx} = 0$
 $\Rightarrow x = \boxed{}$
 $\frac{d^2f(x)}{dx^2} > 0 \rightarrow \text{min}$
 $< 0 \rightarrow \text{max}$

$$\theta_1 = \underline{C}$$

$$\theta_1 - \alpha \times \frac{\partial}{\partial \theta}(\theta)$$

gradient:

$$h_0(x) = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2$$

Handwritten derivation of the partial derivative of the loss function with respect to θ_j :

$$\begin{aligned} \frac{\partial}{\partial \theta_j} L(\theta) &= \frac{\partial}{\partial \theta_j} \frac{1}{2} (h_{\theta}(x) - y)^2 \\ &= 2 \cdot \frac{1}{2} (h_{\theta}(x) - y) \cdot \frac{\partial}{\partial \theta_j} (h_{\theta}(x) - y) \\ &= (h_{\theta}(x) - y) \cdot \frac{\partial}{\partial \theta_j} \left(\sum_{i=0}^d \theta_i x_i \right) \\ &= (h_{\theta}(x) - y) x_j \end{aligned}$$

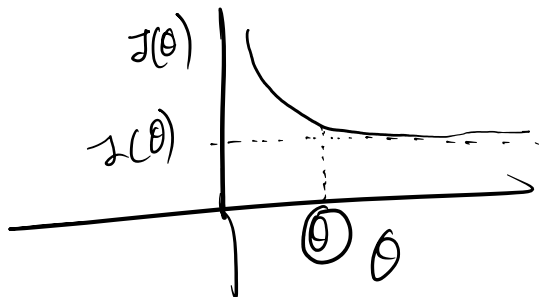
$$J(\theta) = \frac{1}{2} \sum (h_{\theta}(x) - y)^2$$

$$\frac{1}{2} \times 2 (h_{\theta}(x) - y)$$

$$\frac{d}{d\theta}(\theta x)$$

$$\frac{\partial}{\partial \theta_j} \left(\cancel{\theta_0 x_0} + \cancel{\theta_1 x_1} + \dots + \boxed{\theta_j x_j} + \dots \right)$$

$$\frac{\partial}{\partial \theta_j} (\theta_j x_j) = x_j$$



$$\theta = \theta + \sum (y - h(x)) x$$

x_1	y_1
x_2	y_2
$\vdots$	$\vdots$
x_n	y_n

(batch gradient ~~sample~~)
descent

Batch gradient Descent Simulation :

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

$m = 3$ training examples

$n = 2$ features

Training Data :

x_0	x_1	x_2	y
1	1	0	3
1	0	1	4

Iteration 1 initialize:

$$\theta_0 = 0$$

$$\theta_1 = 0$$

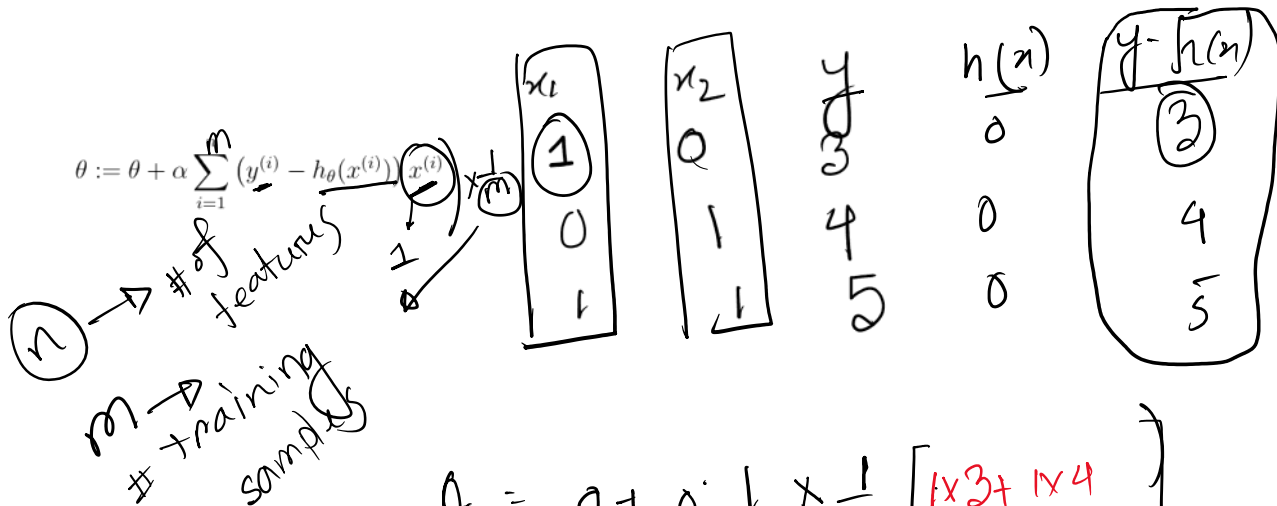
$$\theta_2 = 0$$

$$\alpha = 0.1$$

Step 1

Compute predictions

$$h_{\theta}(x) = 0 \times x_1 + 0 \times x_2 + 0 = 0$$



$$\theta_0 = 0 + \underset{\alpha}{0.1} \times \frac{1}{3} [1 \times 3 + 1 \times 4 + 1 \times 5]$$

$$= 0.4$$

$$\theta_1 = 0 + 0.1 \times \frac{1}{3} [3 \times 1 + (4 \times 0) + (5 \times 1)]$$

$$= 0.267$$

$$\theta_2 = 0 + 0.1 \times \frac{1}{3} [0 \times 3 + 4 \times 1 + 5 \times 1]$$

$$= \underline{\underline{0.3}}$$

$$\theta = (0.4, 0.267, 0.3)$$

$\uparrow \quad \quad \uparrow \quad \quad \uparrow$
 $\theta_0 \quad \theta_1 \quad \theta_2$

$$\frac{1}{2m} (h - y)$$

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x_i) - y(i))^2$$

$$h_1 = \cancel{0.4} \quad 0.4 + 0.267 \times 1 + 0.3 \times 0$$

$$=$$